MILITARY
SPECIFICATION

MIL-C-14468(SigC)
4 February 1957

CONVERTER, TELEPHONE SIGNAL (CONVERTER TA-187( )/U)

1. SCOPE

1.1 This specification covers one plug supervisory type of lightweight, voice frequency ringing equipment designed for field use with Manual Telephone Switchboard SB-86/P or at Telephone Control Office Set AN/TTC-7 equipped with Plug Supervision Trunk TA-276/TTC.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, and standards of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621	Boxes; Wood, Nailed and Locked Corner.
UU-T-111	Tape, Paper Gummed, (Sealing and Securing).
LLL-B-631	Boxes, Fiber, Corrugated (For Domestic Shipment).

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MIL-P-11268	Parts, Materials, and Process Used in Signal Corps Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-S-901	Shockproof Equipment, Class HI (High-Impact) Shipboard Application, Tests for.

STANDARDS

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MIL-STD-105 (With Appendix)	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Inspection Terms and Definitions.
MIL-STD-169	Extreme - Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects For.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions may be obtained from or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.

FSC 5805

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3. REQUIREMENTS

3.1 Description.- Converter, Telephone Signal TA-187( )/U, converts D. C. supervisory signals from Manual Telephone Switchboard SB-36/P or Telephone Central Office Set AN/TTC-7 or their equivalent to 2400 or 2600 cycle per second signals for application to 4 wire trunk circuits and converts 2600 or 2400 cycle per second tone signals to D.C. supervisory signals for application to 2 wire switchboard circuits of Manual Telephone Switchboard SB-86/P and of Telephone Central Office AN/TTC-7( ).

3.2 Parts, materials, and processes; general.- In addition to any requirements of this specification, covering parts, materials, and processes, such items shall conform to Specification MIL-P-11268 including the selection requirements therein.

3.3 Procurement model.- The procurement model shall be the one which will be available for inspection by prospective bidders and will be loaned to the contractor, in accordance with the invitation for bids. Unless otherwise specified herein or in the invitation for bids, the equipment shall conform to the following:

3.3.1 Construction.- Physical construction of the equipment shall conform to the procurement model.

3.3.2 Features.- The equipment shall incorporate all features of the procurement model.

3.3.3 Verification.- Upon receipt of the procurement model, and with the minimum necessary adjustment and alinement (if any), the contractor shall measure its performance for compliance with specified characteristics of the equipment on contract. If the model does not meet the specified performance characteristics, the contractor shall adjust or aline if for optimum performance and then repeat the measurements. The contractor shall immediately prepare copies of a test report on the measurements, and forward them as required by the contract. In case of conflict between specified performance characteristics and performance of the model the difference shall be discussed between the contractor and the contracting officer's technical representative and the proper performance characteristics shall be established.

3.4 Interchangeability.- Corresponding components, replaceable subassemblies, and replaceable parts on contract shall be physically and functionally interchangeable as units without modification thereof of other items with which the units are used. (See 4.10). When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular subassembly or part is to be considered replaceable, the contracting officer shall be consulted.

3.5 Service conditions.- The equipment shall be capable of meet the following service conditions:

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3.5.1 Equipment operating.- The equipment shall be capable of meeting the requirements of this specification under any probable combination of the following:

(a) Operation. Continuous operation when used in a sheltered installation.

(b) Temperature. Any ambient temperature in the range of -40°F to +132°F.

(c) Elevation. Any elevation up to 10,000 feet above sea level.

(d) Relative humidity. Any relative humidity up to 100 percent.

3.5.2 Equipment non-operating. The equipment shall meet the requirements of 3.5.1 after subjection to anyone or more of the following non-operating conditions:

(a) Temperature. Continuous exposure for 24 hours at +160°F and 24 hours at -80°F.

(b) Elevation. Elevation up to 25,000 feet above sea level.

(c) Relative humidity. Relative humidity of 100%, including condensation caused by temperature changes.

(d) Vibration, bounce, and shock. See 3.6.

3.6 Vibration, bounce and shock.

3.6.1 Vibration.- Except for internal resonance of parts and subassemblies specified herein, the equipment shall have no resonance below 55 cycles per second when shock mounts are blocked or removed. (See 4.7.6).

3.6.2 Bounce and shock.- The equipment shall be capable of withstanding the bounce and shock tests of 4.7.4, 4.7.5, 4.7.7, and 4.7.8 without evidence of degradation of specified performance.

3.7 Functional requirements.- Converter, Telephone Signal TA-187( )/U is principally designed for use with Manual Telephone Switchboard SB-86/P or at Telephone Central Office Set AN/TTC-7 equipped with Plug 6 , rvision Trunk TA-276/TTC. It converts D.C. supervisory signals received over a 2-wire trunk from a local switchboard, or central office set, to 2400 or 2600cps voice frequency tone suitable for transmission over 4-wire facilities (generally carrier derived) to distant offices or switchboards. It also reconverts 2400 or 2600 cps voice frequency supervisory signals received from distant installations to D.C. signals suitable for use at the local switching center.

3.8 Electrical requirements.

3.8.1 D.C. Resistance on transmitter-receiver units before installation in chassis assembly.- (See 4.5.2)

MEASUREMENT		D.C. RESISTANCE, ohms
From Pin	To Pin	Value
1 of P1	3 to P3	4170
1 of P1	Ring-TRS-4W	5000
3 of P3	Tip-TRS-4W	5000
6 of K8	Ring-REC-4W	820
9 of K2	6 of K2	4000
5 of K2	3 of K2	4000
3 of V2	K12	1,000,000
7 of K6	K12	200,000

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3.8.2 Power and voltage measurements at each Transmitter-Receiver unit shall be as follows: (See 4.4.2.1).

MEASUREMENTS		VOLTAGE REQUIREMENT		
From Pin	To Pin	Avg.	Min.	Max.
9 of V1	1 of V1	6.3	6.2	6.4
E14 (Neg.)	Gnd (Pos.)	9	8.5	9.5
E3 (Neg.)	Gnd (Pos.)	60	57	63
E7 (Pos.)	Gnd (Neg.)	170	167	173

3.8.2.1 Power Supply output voltage measurements shall be as follows: (See 4.4.2.2)

NOMINAL	TEST POINT	REQUIREMENTS
- 9 dc	- 9 V	- 9.3 ± 0.5 volts (ripple less than 0.1 volts)
- 60 dc	-60 V	- 60 ± 3 volts (ripple less than 0.5 volts)
+ 170 dc	+ 170 V	+ 170 ± 10 volts (ripple less than 1.0 volts)
6.3 ac	From P-101-13	To P101-14	6.4 ± 0.3 volts

3.8.3 Residual noise requirements.- Under test conditions specified in 4.5.5, measurements shall be as follows:

Noise Level	Residual Noise*
> -45 dbm	< -45 dbm

*Remove the "C" to "GRD" connection before measurement.

3.8.4 Receiving loss.- Under conditions specified in 4.5.5., the output measured at the HYB 2W terminals shall be + 6 dbm ± 0.3 db.

3.8.5 Transmitting loss.- Under test conditions specified in 4.5.6, the output measured at the 4 WIRS terminals shall be + 6 dbm ± 0.3 db.

3.8.6 Hybrid balance.- Under the test conditions specified in 4.4.6, the Hybrid Balance shall be greater than 28 db.

3.8.7 Normal tone level.- Under the test conditions specified in 4.4.7 a reading between -17.7 and -19.3 dbm shall be obtained.

3.8.8 Augmented tone level.- Under test conditions specified in 4.4.8, the reading obtained in 3.8.7 shall be increased 12 ± 1 db.

3.8.9 Tone control circuits.- Under the test conditions specified in 4.5.10, tone should be removed from the TRS 4W terminals and the D.C. voltmeter should read 19.7 ± 1 volt.

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3.8.10 Signal receiver no signal test.- Under the test conditions specified in 4.5.11, the voltmeter shall read 5.7 ± 0.4 V, indicating that the signal receiver circuit is released.

3.8.11 Signal receiver minimum signal test.- Under test conditions specified in 4.5.12, the meter reading shall be the same as that obtained in 3.8.10, indicating a released signal receiver circuit.

3.8.11.1 The meter reading should indicate a change occuring at -35 dbm output of the signal generator or lower, under test conditions specified in 4.5.12.1.

3.8.11.2 Under test conditions specified in 4.4.11.2, the measured signal levels shall not change by more than 0.1 db. from those of 3.8.11.1.

3.8.12 Signal receiver strong signal test.- Under test conditions specified in 4.4.12 the signal receiver shall be operated and VCM shall read zero volts.

3.8.13 Signal receiver amplifier limiting.- Under test conditions specified in 4.4.13, the measured values shall be as listed below:

Signal Generator, Output, dbm	AC Voltage Term. 7-14, Z1
+10 dbm	17.6 ± 2.5 V
0 dbm	18.2 ± 2.3 V
-10 dbm	16.7 ± 2.2 V
-20 dbm	15.2 ± 2.1 V
-30 dbm	11.5 ± 2.6 V

3.8.14 Signal receiver 1000 cps interference.- Under test conditions specified in 4.4.14, the decibel meter shall indicate an interference signal amplitude of -20 dbm ± 3 db.

3.8.14.1 Signal receiver 3500 cps interference.- Under test conditions specified in 4.4.14.1, the interference level shall be -24 dbm ± 5 db.

3.8.14.2 Signal receiver outgoing tone interference.- Under test conditions specified in 4.4.14.2, the meter indication shall be between + 3.0 and -3.0 dbm.

3.8.14.3 Signal receiver 180 cps interference.- Under test conditions specified in 4.4.14.3, the meter shall change from approximately 6 volts to zero volts, indicating that the receiver is operating in the presence of the 180 cps interference signal.

3.8.15 Oscillator waveform.- Under test conditions specified in 4.5.16, the trace should be smooth and symmetrical with no flattening at the top or bottom, and with no abrupt irregularities.

3.8.16 Oscillator frequency.- Under test conditions specified in 4.5.17, the frequencies of all four transmitting-receiver units shall be either 2400 ± 5 cps or 2600 ± 5 cps.

3.8.17 Power source.- One Converter Telephone Signal TA-187( )/U shall be capable of meeting all electrical requirements when operated from 115 or 230 volts ± 10%, 50 or 60 cps ± 3 cps, single phase A.C. source, see 4.6.

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3.8.18 Operational test results shall conform to those listed under Table I (See 4.6.).

Table I - Operational Test

Switchboard Condition	Test Board	T - C DC Volts	C - R DC Volts	Position of Switches
Both Idle	A	-55	0	SW 1 either, SW 2 down SW 3 neutral, SW 4 up
	B	-55	0	Same as Above.
A Plugs	A	-18	-22	SW 1 up, SW 2 up SW 3 up, SW 4 down
	B	-55	-11	SW 1 either, SW 2 down SW 3 Neutral, SW 4 up
B Plugs	A	-18	-55	SW 1 up, SW 2 up
	B	-18	-55	SW 3 up, SW 4 down
Talk off	A	-18	-11	SW 1 down, SW 2 up
	B	-18	-11	SW 3 neutral, SW 4 up
A Disconnects	A	-55	-11	SW 1 either, SW 2 down SW 3 neutral, SW 4 up
	B	-18	0	SW 1 down, SW 2 up SW 3 neutral, SW 4 up
B Temporarily Disconnects	A	-55	0	SW 1 either, SW 2 down SW 3 neutral, SW 4 up
	B	0	0	SW 1 down, SW 2 neutral SW 3 neutral, SW 4 up
B Disconnects	A	-55	0	SW 1 either, SW 2 down
	B	-55	0	SW 3 neutral, SW 4 up

3.9 Preproduction samples.

3.9.1 Quantities.- Two complete equipments shall be furnished for Government inspection and approval. These shall be in addition to preproduction samples of parts, materials, and processes specified by Specification MIL-P-11268.

3.9.2 Compliance with specified requirements.

3.9.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (see 3.9.2.3 for exception.)

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3.9.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (See 3.9.4) preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.9.2.3 When deviation from 3.9.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.9.3 Test data.- The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations, and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like.)

3.9.4 Reference standards.- After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of specified requirements.

3.10 Marking.

3.10.1 General.- Marking shall conform to Specification MIL-M-13231 except that the notation "U.S." shall be replaced with "SIGNAL CORPS, U.S. ARMY" on nameplate data.

3.10.2 Serial numbers.- Serial numbers under Specification MIL-M-13231 are required

3.11 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.1).

3.12 Workmanship.- The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs:

(a) In Specification MIL-P-11268:

General requirements for plastic material and plastic parts.	Flux and cleaning agents for soldering.
Wiring and cabling, including:	Process for soldering.
Slack.	Cleaning of equipment.
Clearance.	Riveting.
Connections, general.	General requirements for securing of parts.
Grounding, general.	Cleaning of welding.
Shielding on wire and cable.	

(b) In this Specification 3.10.2 Serial numbers.

3.10.1 General Marking
3.3.1 Construction

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection, general.

4.1.1 Definitions. - Standard MIL-STD-109 applies for definitions of inspection terms used herein.

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4.1.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection of equipment.
(b) Preproduction inspection of preparation for delivery (See Section 5).
(c) Acceptance inspection of equipment.
(d) Acceptance inspection of preparation for delivery (See Section 5).

4.2 Preproduction inspection of equipment.- This inspection shall consist of the group A inspection and nondestructive group B inspection specified in tables III and IV, respectively, and the preproduction inspection specified in table II. Other nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements. Preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, benchhandling, and (4) shock, ballistic or drop; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table II - Preproduction inspection

Inspection (For additional preproduction inspection See. 4.2)	Rqt Par	Insp Par
Temperature test for equipment	3.5	4.7.2
Vibration	3.6	4.7.6
Altitude	3.5	4.7.1
Humidity	3.5	4.7.3
Bounce	3.5	4.7.4
Shock test; ballistic	3.6	4.7.5
Shock test; drop	3.6	4.7.7
Shock test; bench-handling	3.6	4.7.8

4.2.1 Preproduction inspection of preparation for delivery.- This inspection shall consist of the group B inspection specified in table III, and the rough handling test specified in 4.3.4.3.

4.3 Acceptance inspection of equipment.- This inspection shall include group A and group B inspection as specified in 4.3.1 through 4.3.2.3. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B inspection.

4.3.1 Group A inspection of equipment.- This inspection (including sampling) shall conform to table III and Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

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Table III - Group A Inspection of Equipment

Inspection	Rqt Par	Insp Par.	AQL Major	AQL Minor	Control
Visual and Mechanical	3.12	4.11	1.5 dphu	6.5 dphu	
Electrical tests		3.8 thru 3.8.18			
D.C. Resistance - Transmitter Receiver Units	3.8.1	4.5.2 4.1	4.0 dphu for the group		
Power and Voltage	3.8.2	4.4.2			
Residual Noise	3.8.3	4.4.3			
Receiving Loss	3.8.4	4.4.4			
Transmitting Loss	3.8.5	4.4.5			
Hybrid Balance	3.8.6	4.4.6			
Normal Tone Level	3.8.7	4.4.7			
Augmented Tone Level	3.8.8	4.4.8			
Tone Control Circuits	3.8.9	4.4.9			
Signal Receiver No Signal Test	3.8.10	4.4.10			
Signal Receiver Minimum Signal Test	3.8.11 thru 3.8.11.2	4.4.11 thru 4.4.11.2			
Signal Receiver Strong Signal	3.8.12	4.4.12			
Signal Receiver Amplifier L Limiting	3.8.13	4.4.13			
Signal Receiver 1000 cps Interference	3.8.14	4.4.14			
Signal Receiver 3500 cps Interference	3.8.14.1	4.4.14.1			
Signal Receiver Outgoing Tone Interference	3.8.14.2	4.4.14.2			
Signal Receiver 180 cps Interference	3.8.14.3	4.4.14.3			
Oscillator Waveform	3.8.15	4.4.15			
Oscillator Frequency	3.8.16	4.4.16			
Power Source	3.8.17	4.5			
Operational Tests	3.8.18	4.6			1%
All electrical tests are considered major.					

4.3.2 Group B inspection of equipment.- This inspection (including sampling) shall conform to table IV and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of rejected product (sample units and inspection lots) shall be in accordance with Standard MIL-STD-105 and 4.10.

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4.3.2.1 Group B sampling plans.- The group B sampling plans, as listed in table IV shall be as follows:

Group B Plan	AQL	Inspection level for Normal inspection	Inspection level for Reduced inspection
B-2	6.5%	L-7	L-5

4.3.2.2 Procedure in case of failure.- When an inspection lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

Table IV - Group B Inspection of Equipment

Inspection	Rqt Par	Insp Par	Group B Plans
Interchangeability	3.4	4.10	B-2

4.3.3 Reinspection of nonconforming group B sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B inspection may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.4 Acceptance inspection of preparation for delivery.- This inspection shall include group B inspection as specified in 4.3.4.1 through 4.3.4.1.2; no group A or group C inspection is required.

4.3.4.1 Group B inspection of preparation for delivery.- This inspection (including sampling) shall conform to table XXVIII and to the appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1.

4.3.4.1.1 Inspection lot.- A lot for inspection of packaging shall be all unit packages of a particular method or similar combination of methods, submitted at one time. A lot for visual inspection of the pack shall be all completed packs which are identical and are submitted for inspection at one time.

4.3.4.1.2 Procedure in case of failure.- If an inspection lot is rejected the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

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Table V - Group B inspection of preparation for delivery

Inspection	Rqt Par	Insp Par	AQL
Packaging			
Visual inspection, minor --	---		10%
Visual inspection, major --	---		
Heat seal ----------	5.4		4%
Waterproofness ----------	5.4		
Packing			
Visual inspection, minor --	---		10%
Visual inspection, major --	---		4%

4.3.4.2 Conforming group B sample units.- Sample units which have been inspected for preparation for delivery shall be repackaged and repacked to conform to the requirements of section 3.

4.3.4.2.1 Disposition of nonconforming product.- Disposition of lots and sample units rejected during group A or B inspection shall be in accordance with Standard MIL-STD-105. The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

Sample units found defective during group A, B or C inspection.
Inspection lots rejected during group A or B inspection.
Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.3.4.2.2 Disposition of nonconforming product.- Disposition of rejected lots and defective sample units shall be in accordan . with Standard MIL-STD-105. When resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

4.3.4.2.3 Visual inspection of preparation for delivery.- Packages and shipping containers shall be examined for the defects listed in Drawing SC-A-46406, as applicable, and to determine conformance with process sheet specified in 5.5.

Table XXX - Package Tests

Packaging Method	Applicable tests in Specification MIL-P-116
IC-3 - - - - - - -	Waterproofness. Heat seal.
IC-5 - - - - - - -	Water spray. These packages need not be subjected to the water spray test when it will be performed as part of the rough handling test of the pack.
III - - - - - - - -	None.

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4.3.4.3 Rough handling (pilot pack only).- Each completed pack (of the pilot pack, as prepared for shipment) shall be subjected to the rough handling test specified in Specification MIL-P-116. When the pack contains method IC-5 packages, exclusively, the rough handling test shall be followed by the water spray test, as specified in Specification MIL-P-116. Packs shall be opened after completion of the testing and the packages and the items therein examined for damage. These items shall then be subjected to the following inspection:

Operational test paragraph 4.6.

4.4 Electrical tests.- The equipment shall be inspected and tested to determine compliance with the requirements of 3.8 through 3.8.18.

4.4.1 D.C. Resistance.- Test shall be made on all transmitter-receiver units to check that correct resistances are in place, not their accuracy. The D.C. resistance measurements shall meet the requirements of 3.8.1.

4.4.2 Power and voltage checks.

4.4.2.1 With the TA-187/U connected to a power source in accordance with 3.8.17 measurements shall be made as specified in 3.8.2.1 to determine correct voltage input at the transmitter-receiver units as required in 3.8.2.1. All fourm transmitter-receiver units must be installed in chassis before making this test in order to provide the correct load for the rectifier circuits.

4.4.2.2 In making a check of power supply output voltages, at the designated chassis test points, measure all voltages to "C" when "C" is ungrounded. If "C" and chassis GND are connected, either point may be used. The measurements shall be made using a 20,000 ohm per volt multimeter for D.C. voltages and a vaccum tube voltmeter for AC voltages and ripple content. The voltages shall meet the requirements of 3.8.2.2.

4.4.3 Residual noise.- Remove both oscillator tubes from the oscillator and D.C. Power Unit. Connect the "C" binding post to "GND" with a decibel meter terminated in 600 ohms, connected directly across the REC 4W terminals on the transmitter-receiver units, the measurements shall meet the requirements of 3.8.3.

4.4.4 Receiving loss.- With equipment power off and TRS 4W terminals terminated with a 600 ohm ± 5% resistor connect a signal generator set at 1000 cps and 0 dbm output to the REC 4W terminals of the transmitter-receiver units, the measurements shall meet the requirements of 3.8.4. This measurement shall be made with a decibel meter terminated in 600 ohms ± 10% across the MTB 2W terminals.

4.4.5 Transmitting loss.- With equipment power off, terminate the REC 4W terminals of the transmitter-receiver units with a 600 ohm ± 5% resistor. Connect a signal generator set at 1000 cps and 0 dbm output to the MTB 2W terminals and the measurements shall meet the requirements of 3.8.5. This measurement shall be made with a decibel meter terminated in 600 ohms ± 10% across the 4W TRS terminals.

4.4.6 Hybrid balance.- With equipment power off, apply 0 dbm of 1000 cps tone to REC 4W terminals of transmitter-receiver units, measure the output at TRS 4W terminals with no termination on the MTB 2W terminals and also with 600 ohm ± 5% terminations. The hybrid balance shall be the difference between the two readings and shall meet the requirements of 3.8.6.

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4.4.7 Normal tone level.- With equipment power on, adjust the 2400 Output or 2600 Output oscillator control to obtain readings of 1.15 ± .05 V AC from either TST 2400 or TST 2600 terminals to ground, located on the oscillator chassis. Connect a VTVM terminated in 600 ohms ± 10% across the TRS 4W terminals. The measurement shall meet the requirements of 3.8.7.

4.4.8 Augmented tone level.- With equipment power off, ground terminal 2 of capacitor C2A in the Transmitter-Receiver Unit under test. Reapply equipment power and with the same test set-up as in 4.4.7, measure the tone level at the TRS 4W terminals. The measurement shall meet the requirement of 3.8.8. When test is completed turn power off and remove ground from C2A.

4.4.9 Tone Control Circuits.- Connect the "T", "R", and "C" binding posts on the converter to the special test circuit shown in figure 1. Turn equipment power on. Operate key S1 to the Tone Off position ant the voltmeter switch S2 to the Vct position. Take reading on DC voltmeter (Fig. 1). The measurement shall meet the requirement of 3.8.9. Return S1 to the Tone On position and turn signal converter power off to release all equipment relays.

4.4.10 Signal Receiver No Signal Test.- Reapply equipment power with "C" ungrounded and with no signal at the REC 4W terminals, operate the D.C. voltmeter switch S2 to Vcr position. The voltage measurement shall meet the requirements of 3.8.10.

4.4.11 Signal Receiver Minimum Signal Test.- Apply the output of an SG-15 or equal set for minimum output and either 2400 or 2600 cps, depending or the unit being tested to the REC 4W terminals of the transmitter-receiver units, with the DC voltmeter switch S2 set to Vcr, the meter reading should meet the requirements of 3.8.11.

4.4.11.1 Slowly increase the power input to the REC 4W terminals and record the indicated signal generator power at which the meter reading changes from approximately 6 volts to zero volts. The change should meet the requirements of 3.8.11.1. This test checks relay K7 in the converter circuit. Relay K8 should operate within ± 2 db of the signal generator power of the operate value of K7 observed above. Relay K8 is operated when continuity is established from terminal 6 on Transformer T1 to Terminal 3 on relay K8.

4.4.11.2 Perform the same tests outlined in 4.4.11 and 4.4.11.1 above, except that "C" terminal shall be connected to the chassis ground. The measured signal levels shall meet the requirements of 3.8.11.2.

4.4.12 Signal Receiver Strong Signal Test.- This test shall be performed with Terminal "C" connected to chassis ground. With the same connections as outlined in 4.4.11 and 4.11.1 above apply A + 10 dbm signal of the proper test frequency either 2400 or 2600 cps, to the REC 4W input. The measurements shall meet the requirements of 3.8.12.

4.4.13 Signal Receiver Amplifier Limiting.- This test shall be performed with Terminal "C" connected to chassis ground. Apply signal inputs of the appropriate tone frequency in 10 db steps to the REC 4W terminals and measure the AC voltage between terminals 7 and 14 of network Z1. Use a high impedance one-to-one isolation transformer between the network and the meter to avoid grounding the network through the meter. The measured valves should meet the requirements of 3.8.13.

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4.4.14 Signal Receiver 1000 cps interference.- Terminal "C" shall be connected to chassis ground during this test. Connect the T, R and C binding posts as shown in figure 1. Connect the REC 4W terminals to the circuit shown in figure 2. Place switch S3 in the Tone position, switch S4 in the Meas Tone position, and switch S5 in the Meter position. Adjust signal generator #1 to obtain a reading of -14 dbm on the decibel meter, at the appropriate tone frequency. With S3 still in the Tone position, place S4 on Interference and S5 on Test. Adjust the 1000 cps output from signal generator #2 until the DC voltmeter in figure 1 (s2 in the Ver position) changes suddenly from zero to approximately 6 volts. Place S3 in the Meas Interference position and S5 in the Meter position. The resultant meter reading shall meet the requirements of 3.8.14.

4.4.14.1 Signal Receiver 3500 cps interference.- Repeat the test specified in 4.4.15 for an interference signal of 3500 cps. The measured interference level should meet the requirements of 3.8.14.1.

4.4.14.2 Signal receiver outgoing tone interference.- Repeat the test specified in 4.4.15 for an interference signal of the same frequency as the transmitted tone from the unit under test. The measurements should meet the requirements of 3.8.14.2.

4.4.14.3 Signal receiver 180 cps interference.- Place S3 on Meas Interference S4 on Interference and S5 on Meter. Adjust signal generator #2 for a meter reading of -8 dbm. Place S3 on Tone and S4 on MeasTone. Adjust signal generator #1 for a meter reading of -14 dbm. With S3 on Tone, S4 on Interference and S5 on Test, the reading of the DC multimeter, with switch S2 in the Ver position, shall meet the requirements of 3.8.14.3.

4.4.15 Oscillator Wave Form.- Connect a cathode ray oscilloscope across the TST terminals of each oscillator and check for an approximately sinusodial wave form. The shape of the wave forms shall meet the requirements of 3.8.15.

4.4.16 Oscillator frequency.- Connect a frequency counter across the TST terminals of each oscillator. The measurements should meet the requirements of 3.8.16.

4.5 Power Source.- The power source required to operate one Converter Telephone Signal TA-187( )/U, shall meet the requirements of 3.8.17.

4.6 Operational test.- An operational test, meeting the requirements of 3.8.18 (Table I) shall be performed with the use of two test boards, "A" and "B" for systems check in either direction. Test boards shall be constructed using Figure 3 as a guide. For DC volts test from point T to point C and from point C to point R of 2 test boards under conditions specified in table I.

4.7 Service conditions test.- The following service conditions test will be performed on the preproiuction equipments at a Government laboratory. During the service conditions tests, the equipment will be operated continuously or intermittently, as the specification requires, and performance data recorded. Between tests, the equipment shall be checked for such defects as slow dielectric strength, loss of electrical continuity, loose parts, excessive corrosion, breakage, etc. Performance checks shall be made on the equipment at the conclusion of each service condition test. These performance checks shall be sufficiently adequate in number and scope, to uncover any defect resulting from the preceding service conditions tests. Any evidence of degradation of performance below limits specified herein shall be cause for rejection.

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4.7.1 Altitude test.- The equipment shall be operated in a transmit position in a vacuum chamber which has been evacuated to simulate an altitude of 10,000 feet. After one minute under these conditions, the equipment shall be returned to normal atmospheric pressure within five minutes. There shall be no mechanical or electrical failures as a result of this test.

4.7.2 Temperature test.- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169 and shall meet the requirements of 3.8 through 3.8.18.

4.7.3 Moisture-resistance test for equipment.- The equipment shall be tested as follows:

4.7.3.1 Test conditions.-

(a) Do not remove equipment from humidity chamber for measurements.
(b) Start measurements more than 5 minutes after power is applied to the equipment. Complete test measurements as rapidly as possible. Disconnect from power source when measurements are complete.

4.7.3.2 Test method.-

(a) Dry at 130° ± 5°F for 24 hours.
(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.
(c) Test in accordance with 4.5.
(d) Subject to continuous cycling for a total of five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. Repeat test of 4.5.
(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F at 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. The equipment shall meet the full specification requirements of performance. At the completion of the test, the equipment shall be operable and meet the tests of 4.4.

4.7.3.3 Performance.- The performance and ity measurements taken during test shall be equal to the performance before the humidity cycles.

4.7.4 Bounce test.- The unit shall be secured to the adapter plate. The adapter plate shall be placed on the table of the package tester, both as made by L.A.B. Corp., Summit, N.J., or equal. The plate may be loosely constrained in its position by wood cleats. The package tester, shafts in phase, shall be run at a speed of 285 rpm plus or minus 1 percent for 3 hours. The plate shall be rotated 90 degrees every 3/4 hour. The equipment shall evidence no degradation in performance electrically or mechanically as a result of these tests. Superficial surface abrasion is permissible.

4.7.5 Shock test; ballistic.- The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" as shown in Specification MIL-S-901. The equipment shall be secured in its normal operating position to the steel test plate shock testing machine. The test shall consist of a total of 9 blows, one each. 1 foot blow, 3-foot blow and 5-foot blow on the back, side and top of the test plate, respectively. The equipment shall evidence no degradation in performance electrically or mechanically, as a result of these tests. Local mechanical damage consistent with the preceding is permissible.

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4.7.6 Vibration test.- The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.6.1, as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shocks mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration. (See 4.7.6(d).)

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of sub-assemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.7.7 Shock test; drop.- The equipment shall be dropped a distance of one foot on edges and surfaces for a total of 26 drops. Drops may be made from a quick release hook, by and Acme Drop Tester, by a 10° Inclined Plane (Conbur) Tester or by a basket type pendulum tester. The barrier or floor receiving the impact shall be 2 inch fir backed by concrete or by a rigid steel frame. The equipment shall be operable with no serious degradation in performance at the end of this test.

4.7.8 Shock test; bench-handling.- The chassis-and-front panel assembly shall be removed from its inclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as a pivot, and permit the assembly to drop back freely to the horizontal, repeat, using other practicable edges of the same horizontal face as pivots, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly could be placed practicably during servicing.

At the conclusion of this test the equipment shall be operable with no degradation in performance.

4.8 System test.- Converter, Telephone Signal TA-187( )/U shall be tested with companion equipments designated in 3.1 for satisfactory normal traffic channel operation.

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4.9 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.4. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of fasteners on separable parts or assemblies which must mate together, such as covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major components.

(e) Location of connectors, locking pins, fasteners, slides, and mounting on plug-in assemblies.

(f) Size and form of special threads.

4.10 Disposition of nonconforming product.- When defective sample units or rejected inspection lots are resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

4.11 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.- Converter, Telephone Signal (Converter TA-187( )/U) shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116, and described in the following paragraphs.

5.1.1.1 Technical manual.- Each technical manual shall be packaged Method IC-3 as follows: Inclose each technical manual within a close-fitting Type I Class b bag, conforming to Specification MIL-B-117. Heat seal to effect closure.

5.1.1.2 Tubes.- All operating tubes over four inches in length not securely clamped in place, shall be removed, and together with spare tubes, determined in quantity and number from the bid request or contract, packaged individually in conformance with the applicable provisions of Specification MIL-P-75.

5.1.1.3 Accessories and spare parts.- Accessories and spare parts items determined in quantity and number from the bid request or contract shall be packaged individually in accordance with the designated methods as prescribed in Specification MIL-P-116.

5.1.1.4 Converter TA-187( )/U.-

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5.1.1.4.1 Stowage within transit case.- Stow the Converter TS-187( )/U and the items packaged as specified in 5.1.1.1 through 5.1.1.3 together within the compartments of the transit case. Fill all voids with pads fabricated of Type III single faced, flexible, corrugated fiberboard conforming to Specification LLL-F-291. Close the case and secure all fastenings.

5.1.1.4.2 Transit case with contents.- Each transit case with contents shall be packaged Method IC-5 as follows: Cushion each case with cells or pads or both fabricated of double faced corrugated fiberboard having a minimum dry bursting strength of 200 pounds, and designed to absorb the shock of impact in handling and transit. Place the cushioned case within a close-fitting regular slotted Style Grade 5, corrugated fiberboard box conforming to Specification JAN-P-108. Seal the box as specified in the appendix of the box specification and in addition seal all joints and seams with type I, Class 3 tape conforming to Specification PPP-T-60.

5.1.2 Level C.- Converter, Telephone Signal (Converter TA-187( )/U) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.- Converter, Telephone Signal (Converter, TA-187( )/U) shall be packed for shipment, one each within a close-fitting overseas type Style 4 nailed wood box Class 2 conforming to Specification PPP-B-621. Box closure shall be in conformance with the box specification.

5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the container specification only for direct shipment to ports.

5.2.2 Level B.- Converter, Telephone Signal (Converter TA-187( )/U) shall be packed for shipment as specified in 5.2.1 except that the box shall be of the domestic type, Class 1.

5.2.3 Level C.- Converter, Telephone Signal (Converter TA-187( )/U) shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation employed at lowest rates.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

5.4 Package testing.- Packages prepared as above shall be required to pass the preproduction tests and production tests specified in section 4.

5.5 Pilot pack.- Unless otherwise specified in the invitation for bid or contract, a pilot pack shall be furnished for Government inspection and approval. A pilot pack shall consist of all preservation, packaging, packing and marking for delivery for the complete equipment on contract or order. A pilot pack process sheet, using the format shown on Drawing SC-C-33073 shall be prepared by the contractor and forwarded to the contracting officer and shall be accompanied by such drawings, sketches and other data as may be necessary to completely describe the procedure followed in fabricating the pilot pack and to identify the equipment items therein.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (Level A, level B, or level C.)

(c) Samples required. (See 3.9)

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required. (See 3.11.)

(g) Quantity of tools and running spare parts required. (See 3.11.)

6.2 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example TA-187W. As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation, whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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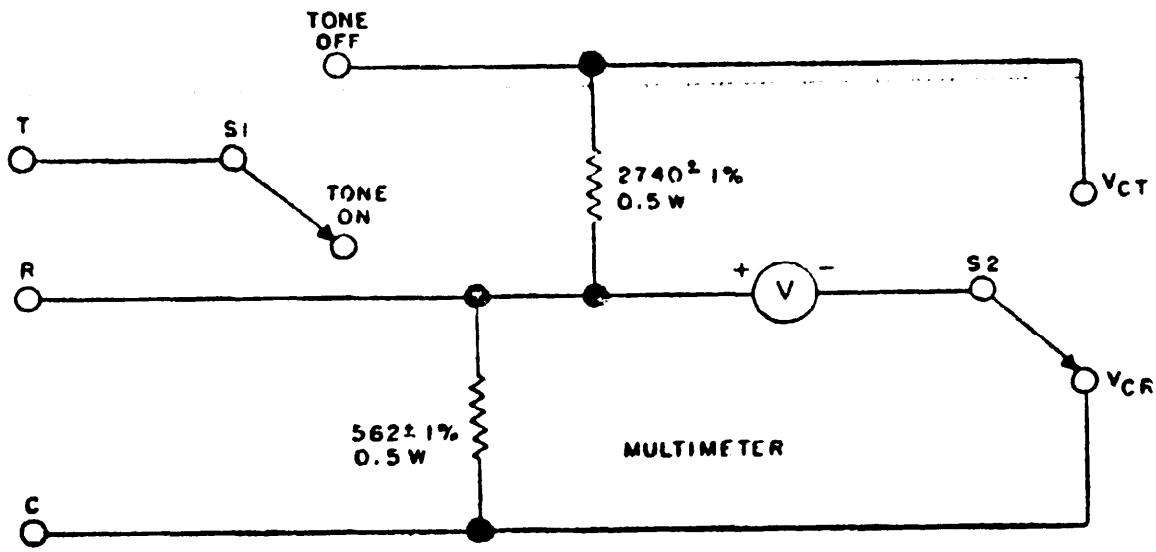


FIG.1 CONVERTER, TELEPHONE SIGNAL TA-187( )/U TEST SET FOR TONE CONTROL CIRCUITS

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SPECIFICATION

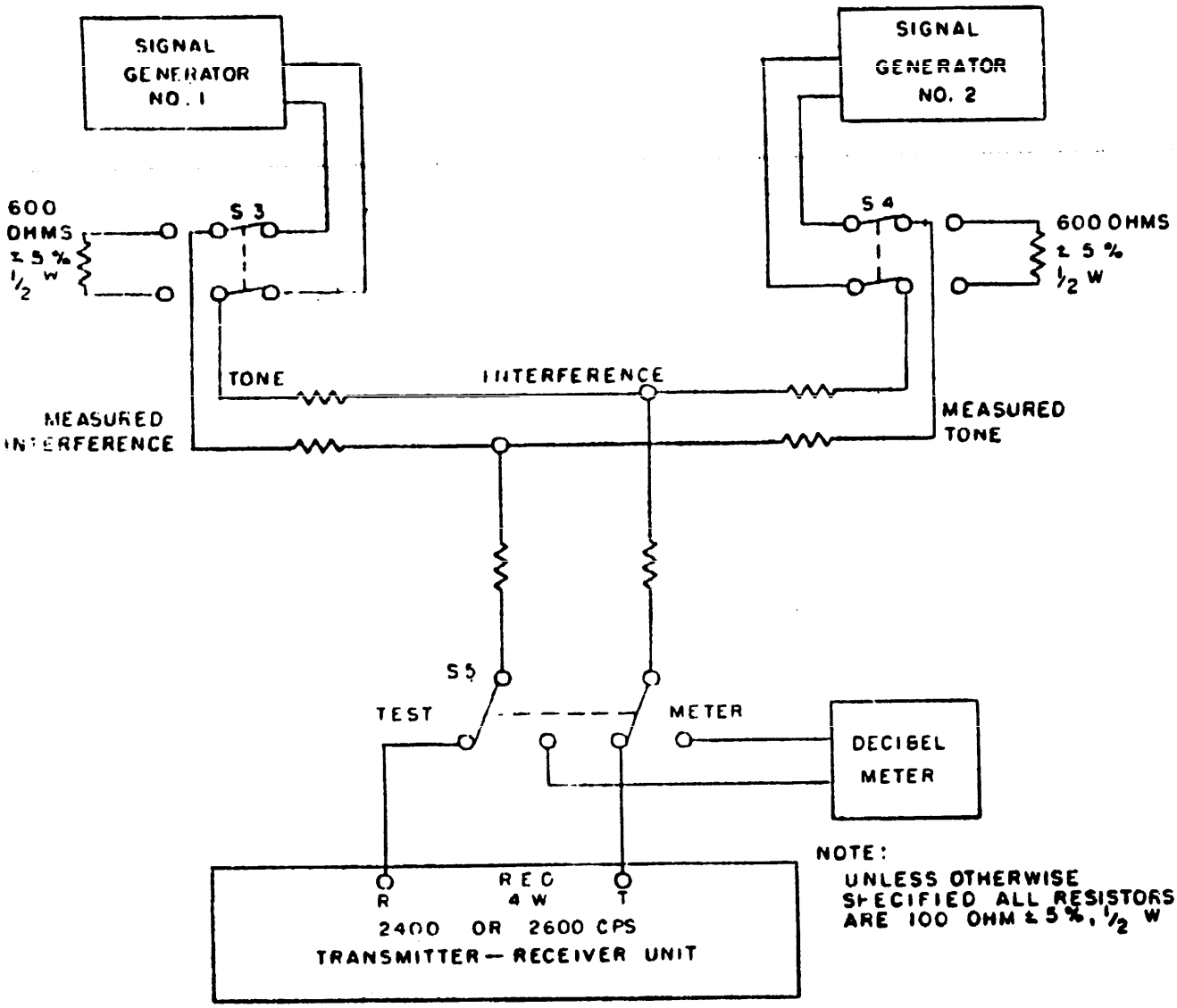
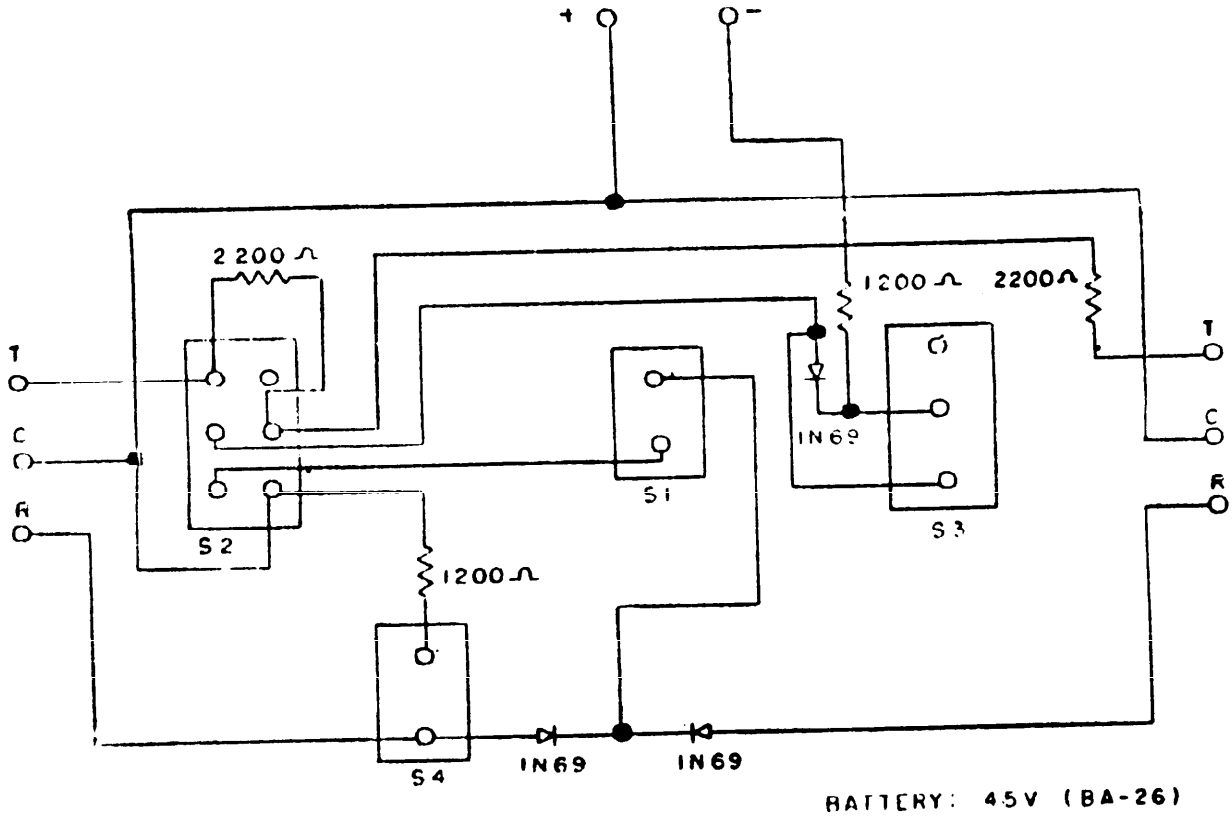


FIG 2. CONVERTER TELEPHONE SIGNAL TA-187( )/U

TEST CIRCUIT FOR INTERFERENCE TESTS

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BATTERY: 45V (BA-26)

SWITCHES:

SW 1 - SPST
SW 2 - DPDT
SW 3 - SPDT
SW 4 - SPST

FIG. 3 TEST CIRCUIT FOR OPERATIONAL TEST

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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